

Homework A1 | ECON 0100 | Fall 2026

Due: Sunday, Aug. 30

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely, and show your work so you can later understand your thought process. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

The Hogsmeade Candy Shop Saga

Honeydukes is a candy shop in Hogsmeade. The shop makes two types of popular candy, Chocolate Frogs (F) and Ice Mice (M). Devoting all their resources to one good, Honeydukes can make 100 pounds of F or 75 pounds of M each week.

Q1 | Honeydukes' PPF

a) Set up Honeydukes' weekly PPF on an x, y graph with Ice Mice (M) on the vertical and Chocolate Frogs (F) on the horizontal.

b) What is Honeydukes' opportunity cost of 1F?

Q2 | Feasibility

The shop receives three owl-orders for next week. Classify each production plan as inefficient, efficient, or unattainable, using your graph or algebra to justify your answers.

a) 40F and 30M

b) 60F and 30M

c) 80F and 30M

Q3 | Dynamics

Ahead of the holiday rush, Honeydukes hires a second shift, doubling the shop's labor. Right after, their Chocolate Frog mold cracks, cutting their Chocolate Frog productivity in half.

- a) Show both changes to Honeydukes' PPF on one graph.
- b) After both changes, what is Honeydukes' opportunity cost of 1F? Of 1M?

Q4 | Next Best Alternative

Neville visits Honeydukes with one galleon, enough to buy exactly one item: a pound of Chocolate Frogs, a pound of Ice Mice, or a Sugar Quill. He likes Chocolate Frogs best, then Sugar Quills, then Ice Mice.

- a) What is Neville's opportunity cost of buying the Chocolate Frogs?
- b) Honeydukes adds Fizzing Whizzbees to the shelf, also one galleon. Neville likes them more than Sugar Quills but less than Chocolate Frogs. What is his opportunity cost of the Chocolate Frogs now?

<!-- The questions are free response on paper; students submit selections on Gradescope. The selection form and answer key live in Homework_A1_gradescope.md. ANSWER KEY (not for students) Q1a: $M = 75 - (3/4)F$ Q1b: $3/4$ pound of M Q2a: Inefficient Q2b: Efficient Q2c: Unattainable Q3a: doubling labor shifts out to 200F or 150M; the cracked mold pivots F in to 100F or 150M Q3b: OC of 1F = $3/2$ M; OC of 1M = $2/3$ F Q4a: A Sugar Quill Q4b: A Fizzing Whizzbee -->